

Practical Fixed-Income Demos

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These executable demonstrations connect the lecture models to realistic analyst and portfolio-management workflows. Each case separates observed or supplied inputs, estimated quantities, diagnostics, and the decision that uses the output. The curve and Vasicek cases embed dated official market observations for deterministic offline rendering; other cases clearly identify their market-like teaching inputs. HTML code is folded by default, while results and mathematical conventions remain visible.

Lecture	Workflow	Demo
3	Duration, DV01, convexity, and full-repricing comparisons	Duration and convexity workflow
6	Benchmark-aware sampling, constrained optimization, trades, and stress tests	Bond portfolio construction workflow
7	Dated Treasury curve, bootstrapping, Hull–White calibration, and future valuation	From observed yields to Hull–White future yield curves
7	Effective-fed-funds estimation, Vasicek scenarios, and funding-risk application	Vasicek scenario workflow
8	Callable-bond valuation, option cost, OAS, and effective risk	Callable-bond and OAS workflow

	Lecture	Workflow	Demo
	9	DV01-based Treasury futures hedging and residual risks	Treasury futures hedging workflow
	10	Spread duration, DTS, carry, expected loss, and stress tests	Credit-spread risk workflow
	11	Treasury futures options and asymmetric portfolio protection	Interest-rate options workflow

The examples are teaching tools rather than trading recommendations. Production valuation requires current market conventions, instrument terms, liquidity inputs, and independent model validation.